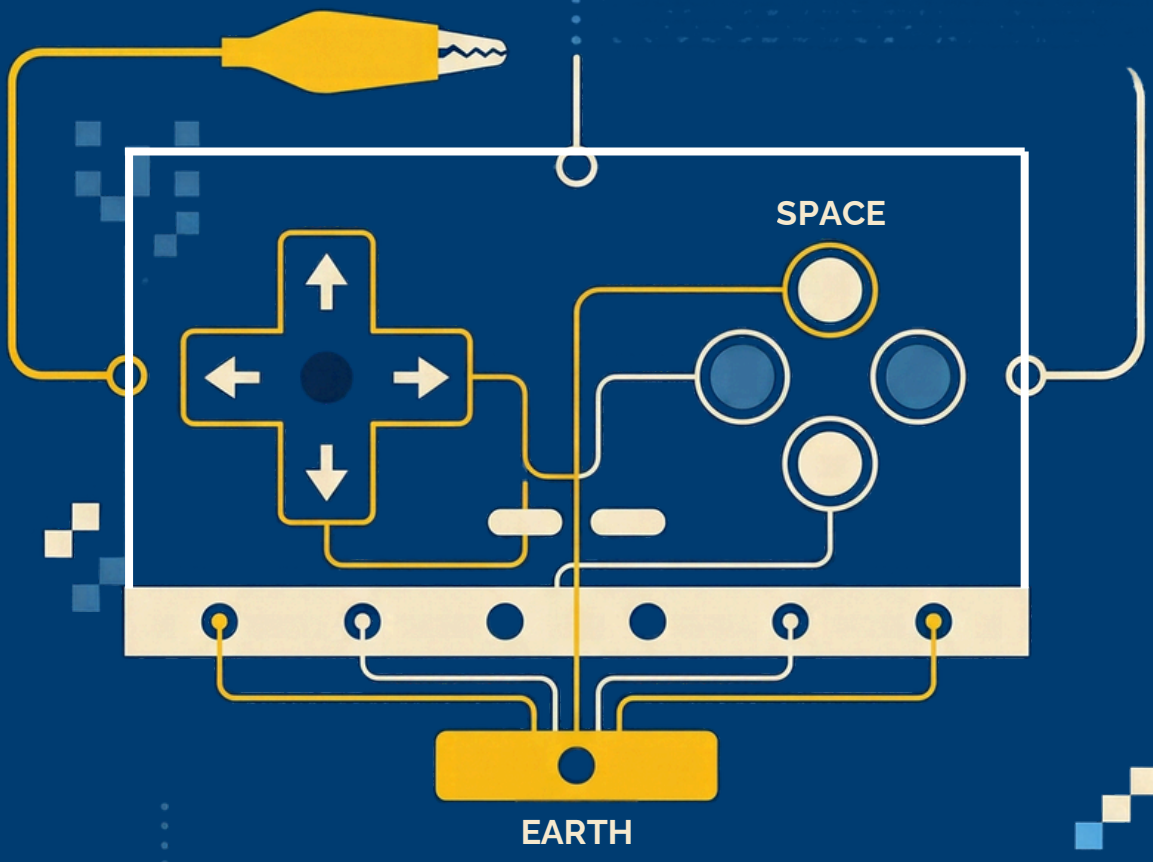
STEMVERSE

STEMverse

GAME DEVELOPER KIT

PLAY. BUILD. INTERACT.

TURN THE PHYSICAL WORLD INTO YOUR GAME CONTROLLER.



Turn the physical world into your game controller.

WELCOME TO THE GAME STUDIO

Hello, Game Developer!

You are holding a powerful tool used by inventors around the world. The STEMverse Game Developer Kit lets you trick your computer into thinking almost *anything* is a keyboard.

Bananas? Yes. Play-Doh? Yes. Aluminium tape? Definitely!

In this manual, you will learn how to design physical video game controllers. Instead of just tapping standard plastic buttons, you will build musical instruments, arcade pads, and maze navigators using everyday conductive materials.

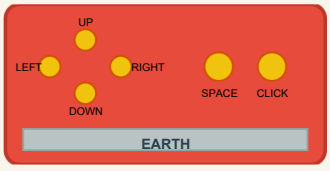
HARDWARE REQUIREMENT:
To use this kit, you MUST have access to a computer or laptop (Windows, Mac, or Chromebook) with a standard USB port. This kit does not have a screen or battery—it is the bridge between the physical world and your computer!

YOUR GAME DEV JOURNEY

- **Learn the Tech:** Discover how the Makey Makey board reads electricity.
- **Build the Templates:** Complete 5 physical game interface missions.
- **Invent the Future:** Design a controller that has never existed before.

WHAT'S IN YOUR KIT?

Beforewestartcodingphysicalinputs,let'scheckyour inventory.



1X MAKEY MAKEY

The brainofthe operation.Itturnselectrical connections into keyboard presses.



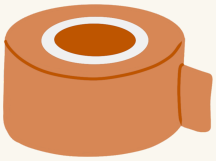
1X USB CABLE

Connects theMakeyMakey toyourcomputer and provides safe power.



6X CROCODILE WIRES

Wires with metal "teeth" that clip onto the board and your conductive materials.



1M ALUMINIUM TAPE

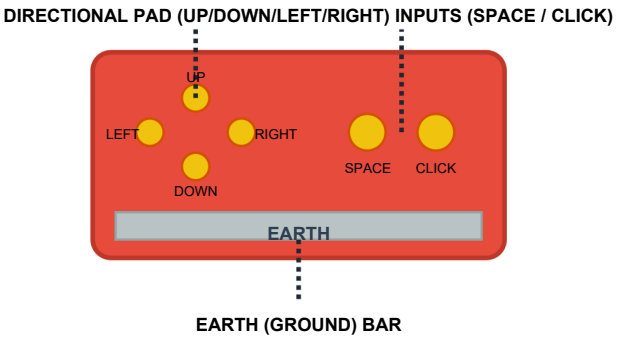
Conductive metal tape. Electricity easily flows right through it!

5X PRINTED GAME TEMPLATES

Five physical game boards you will build and wire up.

Meet your Physical Computing Board

This is a printed circuit board. Let's look at the front layout:



1. THE INPUTS

The holes with metal rings on the front map to exactly what they say: `UP` `DOWN`, `LEFT`, `RIGHT`, `SPACE`, and `CLICK`.

2. EARTH (GROUND)

The long metal strip at the bottom. An electrical circuit is a circle. To make an input work, electricity must flow from EARTH, through you, to the input pad!

HOW DOES IT WORK?

The magic of physical computing comes down to **Conductivity** and **Circuits**.

STEP 1: THE OPEN CIRCUIT

When you plug the Makey Makey into your computer, it sends a tiny, safe amount of electricity to the EARTH bar. But the electricity is trapped. It has nowhere to go. This is an *Open Circuit*.

STEP 2: THE HUMAN WIRE

Humans are mostly water, which means **you are conductive!** When you hold a wire connected to EARTH, the electricity travels safely into your hand.

STEP 3: CLOSING THE LOOP

When your other hand touches an input (like the SPACE circle), you create a bridge! The electricity flows out of EARTH, through you, into the SPACE input, and back to the board. The Makey Makey detects this *Closed Circuit* and sends a "Spacebar pressed" message to the computer.

YOUR FIRST CONNECTION

Let's prove that you can become a human wire.



BUILD IT

1. Plug the USB cable into the board, and the other end into your computer. (The board should light up).
2. Open a text editor on your computer (like Notepad, Word, or Google Docs).
3. Take one Crocodile Wire. Clip it to the **EARTH** bar at the bottom of the board.
4. Take a second Crocodile Wire. Clip it to the **SPACE** circle.
5. Hold the metal end of the EARTH wire in your left hand.
6. With your right hand, tap the metal end of the SPACE wire!

WHAT HAPPENED?

Look at your screen! Every time you tapped the wire, a space appeared in your document. You just bypassed your physical keyboard entirely.

UNDERSTANDING INPUTS

Game developersdon'tjustuseone button.Theymapdifferent actions to different inputs. The Makey Makey has standard **Key Mapping** built-in.

Board Label	Computer Output	Common Game Use
UP	Up Arrow Key (↑)	Jump, Move Forward
DOWN	Down Arrow Key (↓)	Duck, Move Back
LEFT	Left Arrow Key (←)	Move Left
RIGHT	Right Arrow Key (→)	Move Right
SPACE	Spacebar	Shoot, Select, Jump

We will use these 5 inputs for all the missions in this workbook.

HOW TO USE THE TEMPLATES

Your kit includes 5 printed Game Templates.These are your physical interface boards! Here is the workflow for every mission:

1. PREPARE THE SURFACE

Lay the paper template flat. Look for the grey dotted zones marked **"PLACE TAPE HERE"**.

2. APPLY ALUMINIUM TAPE

Cut or tear strips of the shiny Aluminium Tape. Peel off the paper backing and stick the tape directly over the marked zones. *Make sure the tape strips don't touch each other!* If two pads touch, the computer will press two buttons at once.

3. CONNECT THE CROCODILE WIRES

Clip one end of a crocodile wire to the edge of the paper so its metal teeth bite down on the aluminium tape. Clip the other end to the matching input on the Makey Makey.

4. ESTABLISH EARTH

Every template requires an EARTH connection. Connect a wire from the boards EARTH bar. You must hold the metal end of this wire (or tape it to a conductive bracelet) while you play.

MISSION 01

PIXEL PIANO

Mission Brief: Standard pianos are boring. Let's invent a paper piano that plays real digital music when you touch the aluminium keys!

Your Toolkit: Template 01 (Pixel Piano), 1x Makey Makey, USB Cable, Aluminium Tape, 6x Crocodile Wires, Computer.

BUILD IT

1. Apply strips of aluminium tape to the 5 designated piano keys on Template 01.
2. Connect 5 crocodile wires from the taped keys to the Makey Makey inputs: UP, DOWN, LEFT, RIGHT, and SPACE.
3. Connect your 6th wire to EARTH.

PLAY IT

On your computer, open a web browser and go to: **makeymakey.com/piano**

Hold the metal end of the EARTH wire in one hand. Use your other hand to tap the aluminium piano keys. The website is programmed to play notes when it detects Arrow keys or Spacebar presses!

LEVEL UP!

Engineer Challenge: Can you play a chord? Try touching the LEFT and RIGHT pads at the exact same time.

Inventor Challenge: Can you redesign the piano so you don't have to hold the EARTH wire? (Hint: Can you make an aluminium tape bracelet?)

MISSION 02

BEAT BUILDER

Mission Brief: Time to make some noise. Wire up your physical drum pads to create a custom electronic drum kit.

Your Toolkit: Template 02 (Beat Builder), Makey Makey, Tape, 6x Wires, Computer.

BUILD IT

1. Apply aluminium tape to the 5 large drum pad targets on Template 02.
2. Wire the pads to the 5 Makey Makey inputs.
3. Connect the final wire to EARTH and hold it.

PLAY IT

On your computer, go to: **makeymakey.com/bongos**

Tap the physical pads. Notice how the computer reacts to the physical connection instantly? This is called low-latency **Human-Computer Interaction (HCI)**.

WHAT'S HAPPENING?

When you map physical touches to digital sounds, you are doing **Interaction Design**. The computer doesn't know it's a drum, it just knows you triggered the "UP" key. The software handles the audio!

Inventor's Log: Write down your 4-beat rhythm pattern here (e.g., Space, Space, Up, Down):

_____

MISSION 03

ARCADE CONTROLLER

Mission Brief: Replace your boring plastic keyboard with a custom retro arcade controller to play classic games.

BUILD IT

1. Apply tape to the 4 Directional buttons and the 1 Action button on Template 03.
2. Wire them accurately: Make sure the physical UP button goes to the Makey Makey UP input! (Proper **Input Mapping**).
3. Connect your EARTH wire.

PLAY IT

Open any web-based game that uses Arrow Keys to move and Space to jump (like Pac-Man clones or platformers on Scratch). Hold EARTH and play the game using your paper controller!

LEVEL UP!

Engineer Challenge: Play the game for 2 minutes. Are the buttons too close together? Did the crocodile clips get in the way of your hand? This is called testing **Ergonomics**.

Inventor's Log: If you built this controller out of cardboard instead of paper, where would you place the wires to keep them hidden?

_____

MISSION 04

QUIZ SHOWDOWN

Mission Brief: It's a two-player battle! Build a game-show buzzer system to see who has the fastest reaction time.

BUILD IT

1. Apply tape to Player 1's buzzer, Player 2's buzzer, and the shared EARTH pad in the middle of Template 04.
2. Wire Player 1 to LEFT. Wire Player 2 to RIGHT.
3. Wire the middle EARTH pad to the Makey Makey EARTH bar.

PLAY IT

Instead of holding the EARTH wire, **both players** must keep one finger resting on the shared EARTH tape pad on the paper. When the host asks a question, the first player to touch their answer buzzer closes the circuit!

WHAT'S HAPPENING?

The Makey Makey only detects the input. It doesn't know who was first. You need software to determine the winner! You can build a simple script in **Scratch (scratch.mit.edu)**: *"If Left Arrow pressed, say Player 1 wins! If Right Arrow pressed, say Player 2 wins!"*

LEVEL UP!

Inventor Challenge: The current design requires players to keep a finger on the EARTH pad. Can you design a physical rule where they have to high-five the host (who is holding the EARTH wire) to answer?

MISSION 05

MAZE MASTER

Mission Brief: Precision navigation. Build a futuristic console controller to steer a digital character throughacomplex maze.

BUILD IT

1. Apply tape to the 4 directional nodes and the center Action node on Template 05.
2. Carefully wire the pads to their corresponding Makey Makey inputs. Keep the wires neat so they don't tangle your hands!
3. Connect EARTH.

PLAY IT

Find a maze game online that uses arrow keys. Practice navigating. Because the aluminium tape is completely flat, you have to rely entirely on muscle memory, as there is no physical "click" feeling (tactile feedback).

WHAT'S HAPPENING?

Tactile Feedback is why standard controllers have physical plastic buttons that click. Without it, your brainhastorelypurelyon thevisualoutputon the screento know if thebutton press worked!

DESIGN YOUR OWN CONTROLLER

You have mastered the templates. Now, it's time to build from scratch.

THE INTERFACE RULES

- You can map a maximum of 5 actions.
- You MUST include a way for the player to connect to EARTH.
- The conductive pads must be easy to reach.

Makey Input	My Game Action (e.g., Jump, Fire)
UP	
DOWN	
LEFT	
RIGHT	
SPACE	

[Sketch Your Physical Layout Here]

GAME DEVELOPER CHALLENGE

PROMPT: You are a lead game developer. You have been asked to design a new way for people to interact with a computer game. You CANNOT use a traditional keyboard or mouse. You CANNOT use paper templates. What will you invent out of everyday household objects?

IDENTIFY THE GAME

What game are you designing for? _____

THE INTERFACE

What everyday conductive objects will become your buttons? (Hint: Bananas, coins, foil-wrapped cardboard, spoons, graphite pencil drawings).

_____

HOW DO YOU PLAY?

Explain how the player uses your physical invention. Where do they put their hands? How do they connect to EARTH?

TROUBLESHOOTING

Engineers don't panic when things break. They debug! Use this guide to fix your broken game controllers.

The Bug	The Fix
Makey Makey not detected.	Check the USB cable. Is it fully inserted into both the board and the computer?
No input registering on computer.	Did you forget EARTH? You MUST be touching the metal end of the EARTH wire for the circuit to close.
Input is registering continuously!	A short circuit! Your EARTH tape pad and your Input tape pad are physically touching on the template. Separate them.
Aluminium tape isn't working.	Ensure the crocodile clip's metal teeth are directly biting the shiny top side of the tape.
Computer app isn't responding.	Click on the web browser window with your mouse first to make sure it is "active" and listening for keyboard inputs.

MY GAME DEVELOPER LOG

My Favorite Invention Name: _____

The smartest thing I designed was:

The hardest bug to fix was:

Next time, I want to improve:

STEMVERSE

GAME DEVELOPER CERTIFICATE

This certifies that

_____

has successfully completed all Game Developer Missions, demonstrating mastery of physical computing, input mapping, interaction design, and hardware troubleshooting.

_____ _____

Date **Lead Developer**

PLAY. BUILD. INTERACT.